

Hip Dysplasia and Instability

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OVERVIEW OF PATHOLOGIES

Hip dysplasia, or developmental dysplasia of the hip (DDH), is a common cause of pain in young adults, including athletes. Hip dysplasia encompasses a wide spectrum of disease pathology, from severe dysplasia with hip subluxation/dislocation, to milder deformities often not recognized until adulthood. Several factors appear to play a role in the pathophysiology of hip dysplasia, including acetabular deformity (acetabular dysplasia), femoral deformity (femoral dysplasia), and soft tissue laxity. A thorough understanding of hip dysplasia is important for all clinicians because the history and physical examination of many patients with hip dysplasia can be very similar to other hip disorders, including femoroacetabular impingement (FAI). Hip dysplasia has a relatively clear natural history with progressive damage to the acetabular rim and eventual osteoarthritis. The periacetabular osteotomy (PAO) has become the accepted standard for treatment of acetabular dysplasia to alter the underlying structural hip instability. Good outcomes of PAO have been reported out to 20 years postoperatively. The PAO has undergone significant evolution in the past two decades to improve the recovery from the procedure, including successful return to sporting activities. Increased recognition of the role of labral and articular cartilage pathology in the dysplastic hip has increased the role of hip arthroscopy in this population. Isolated hip arthroscopy in the dysplastic hip is associated with high rates of failure, including some catastrophic complications, including hip dislocation and rapid progression of osteoarthritis. Hip arthroscopy appears to have a role in combination with PAO in patients with significant labral or cartilage pathology. Optimal acetabular correction and bony correction of any associated head-neck offset deformity appear to be important to avoid creating FAI after PAO. Symptomatic hip instability is increasingly recognized as a cause of hip pain in individuals with borderline hip dysplasia or atraumatic hip microinstability. The role of bony correction versus soft tissue plication remains to be better defined in future research. In absence of bony deformity, capsular plication appears to have a role in addition to treatment of labral pathology.

Hip Dysplasia

Hip dysplasia encompasses a complex and variable group of deformities of the hip. Acetabular dysplasia is the most commonly recognized and treated component of hip dysplasia. Classic

acetabular dysplasia results from insufficiency of the anterosuperior acetabulum. This results in overload of the anterosuperior acetabular rim, which Klaue et al. termed “acetabular rim syndrome.”¹ In the nondysplastic hip, the acetabular labrum functions to seal the hip joint rather than absorb direct load.²⁻⁴ However, in the dysplastic hip, the labrum and acetabular rim are subjected to direct load, which contributes to the typical labrochondral pathology in these locations.⁵ The increased forces seen by the anterosuperior labrum may play a role in the development of labral hypertrophy, which is generally but not uniformly present. The hypertrophic labrum may provide increased hip stability in the setting of acetabular dysplasia but also appears susceptible to ultimate overload and tearing.⁶ The radiographic appearance of acetabular dysplasia is variable (Fig. 81.1).⁷ Severe acetabular dysplasia, with associated hip dislocation or subluxation, is generally well recognized on plain radiographs, with a break in the Shenton line recognized as a simple radiographic marker of hip subluxation. Mild or borderline acetabular dysplasia requires a thorough assessment of hip morphology to avoid missing this diagnosis. The variability of acetabular version in the dysplastic hip and recognition of the variable patterns of deformity are increasingly noted. Nepple et al.⁷ defined three common patterns of acetabular deformity in hip dysplasia based on low-dose computed tomography (CT). Anterosuperior, global, and posterosuperior insufficiency of the acetabulum were all fairly common (Fig. 81.2).

The femoral deformity in hip dysplasia is generally less severe and less commonly treated than the acetabular deformity, although in some cases femoral deformity may predominate.⁸ Increased femoral anteversion is commonly present in the dysplastic hip and places additional stress on the anterior hip stabilizers. Normal femoral anteversion is approximately 5 to 15 degrees. In the setting of hip dysplasia, hips with anteversion more than 20 degrees above normal are relatively common, with severe increases in femoral anteversion greater than 35 degrees being less common. Coxa valga may be present in some dysplastic hips, although plain radiographic assessment of the femoral neck-shaft angle in high with increased femoral anteversion may give the appearance of more severe coxa valga than is truly present. Wells et al. reported the variability of femoral morphology in the dysplastic hip.⁸

The natural history of acetabular dysplasia has been fairly well defined, compared with other young adult hip disorders. However, this natural history is solely based on radiographic

Abstract

Hip dysplasia, or developmental dysplasia of the hip (DDH), is a common cause of pain in young adults, including athletes. Hip dysplasia encompasses a wide spectrum of disease pathology, from severe dysplasia with hip subluxation/dislocation, to milder deformities often not recognized until adulthood. Several factors appear to play a role in the pathophysiology of hip dysplasia, including acetabular deformity (acetabular dysplasia), femoral deformity (femoral dysplasia), and soft tissue laxity. A thorough understanding of hip dysplasia is important for all clinicians because the history and physical examination of many patients with hip dysplasia can be very similar to other hip disorders, including femoroacetabular impingement (FAI).

Keywords

hip dysplasia
acetabular dysplasia
instability
PAO

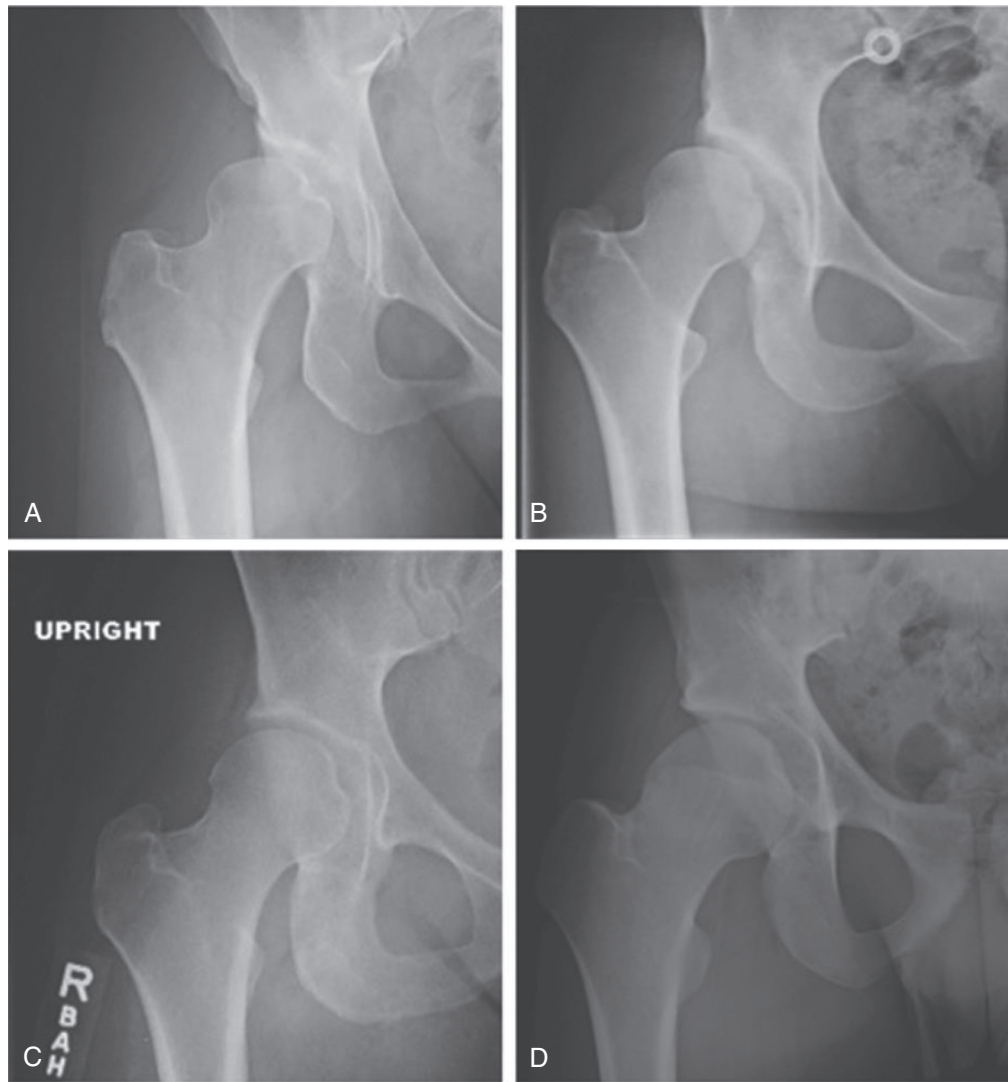


Fig. 81.1 Hip dysplasia variants. (A) Moderate hip dysplasia with associated break in the Shenton line. (B) Moderate hip dysplasia with intact Shenton line. (C) Mild hip dysplasia with normal femoral morphology. (D) Mild hip dysplasia with associated cam morphology of proximal femur. (Reprinted from Nepple JJ, Clohisy JC. The dysplastic and unstable hip: a responsible balance of arthroscopic and open approaches. *Sports Med Arthrosc.* 2015;23[4]:180–186.)

characterization of the lateral center edge angle. The role of other factors, including femoral version, sex, soft tissue laxity, and activity level, remains to be better defined. Wiberg described the lateral center edge angle in his 1939 study on hip dysplasia and suggested that a value less than 20 degrees was linked to osteoarthritis.⁹ He reported a cohort of 18 such patients who all developed osteoarthritis at 4- to 29-year follow-up. This threshold appears to capture the lowest 5% of acetabular coverage within a population. It is important to note that deformities of acetabular dysplasia occur less commonly than those of FAI morphology, which are generally 10% to 25%. Cooperman et al.^{10,11} subsequently demonstrated similar progression of osteoarthritis among a cohort of 20 hips. However, the rates of osteoarthritis development did not appear to correlate with the severity of dysplasia and are likely influenced by a number of other patient-specific factors. In another key study in 1986, Murphy et al.¹² reported the outcomes of the contralateral hip of 286

patients undergoing total hip arthroplasty for hip dysplasia. All hips with an lateral center edge angle (LCEA) less than 16 degrees developed osteoarthritis by the age of 65. However, not all studies have been in agreement on the link between hip dysplasia and osteoarthritis. Jacobsen et al.¹³ followed 81 dysplastic hips over 10 years (compared with controls) and demonstrated no difference in joint space width in hips with dysplasia. Hips with mild dysplasia (LCEA 15 to 25 degrees) did not uniformly develop osteoarthritis during the 10-year period. Lane et al.¹⁴ performed perhaps the best level of evidence in a prospective cohort study. They demonstrated that LCEA less than 30 degrees was associated with a 3.3 times increased risk of osteoarthritis at 8-year follow-up. This supports the potential increased risk for structural instability and hip osteoarthritis in hip with borderline dysplasia (LCEA 20 to 25 degrees). Li et al.¹⁵ demonstrated the potential role of femoral anteversion in the dysplastic hip and found that hip osteoarthritis development was associated with increased

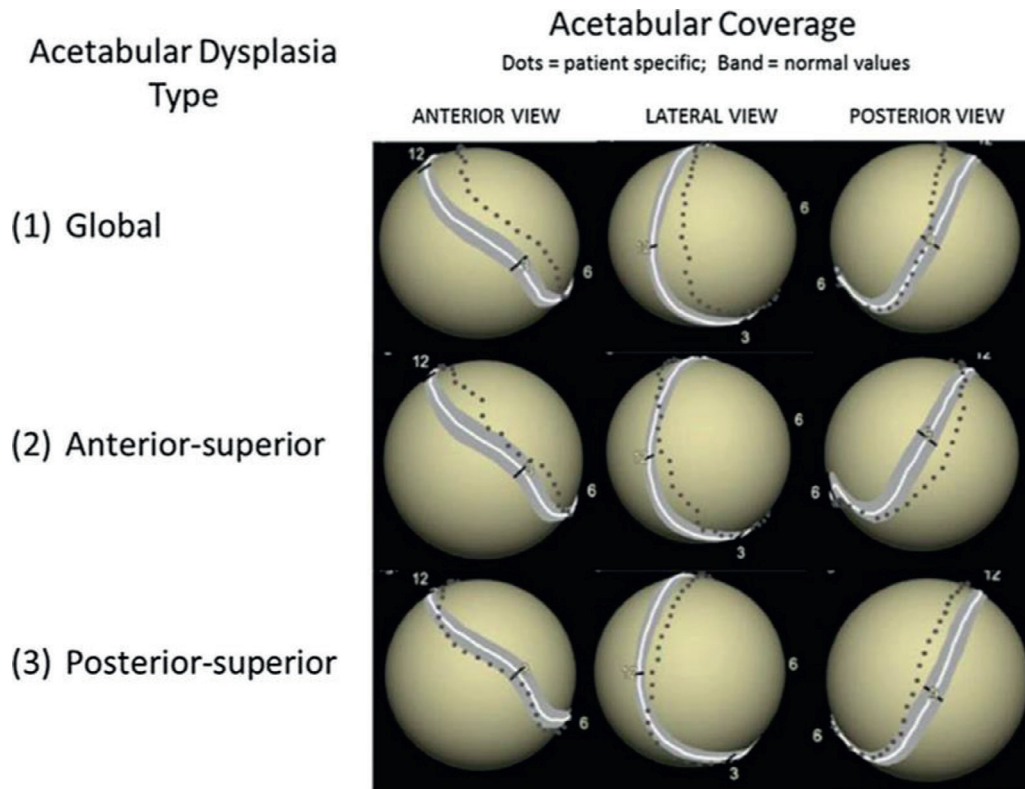


Fig. 81.2 Three subtypes of bony deficiency in the setting of acetabular dysplasia are common, including global, anterosuperior, and posterosuperior deficiency patterns. (Reprinted from Nepple JJ, Wells J, Ross JR, et al. Three patterns of acetabular deficiency are common in young adult patients with acetabular dysplasia. *Clin Orthop Relat Res.* 2017;475[4]:1037–1044. doi:10.1007/s11999-016-5150-3.)

femoral anteversion (osteoarthritis, mean femoral anteversion 18 degrees; compared with 15 degrees with no osteoarthritis). Combined acetabular and femoral anteversion appears to play a similar role. The authors recommended combined anteversion greater than 40 degrees or femoral anteversion greater than 20 degrees as risk factors for hip osteoarthritis. The natural history of dysplasia in males is less clear. Croft et al.¹⁶ found no association of hip dysplasia (LCEA <25 degrees) with osteoarthritis in a study of 1516 pelvic radiographs.

HISTORY

As in all young adult hip disorders, a comprehensive history and physical examination plays an important role in arriving at a proper diagnosis. Patients with symptomatic acetabular dysplasia typically present with groin pain, although lateral pain secondary to abductor overload is also commonly present. In addition, more than 20% report a history of low back pain.¹⁷ One study reported anterior groin pain in 72% of patients with hip dysplasia and lateral hip pain in 66%. Most patients describe activity-related pain with an insidious onset and gradual progression. Most commonly patients report a duration of symptoms of 1 to 3 years prior to any surgical treatment.¹⁷ Pain is often worsened by prolonged walking or other activities, although in some cases it is worsened by sitting. Most patients report no history of childhood hip disease including DDH. A family history of hip dysplasia or osteoarthritis is present in many patients. Higher

activity level and greater dysplasia severity appear to result in pain development at a younger age.¹⁸ Although definitely more common in females (constituting 72% to 83% in most studies),^{17,19} hip dysplasia also occurs in males.

PHYSICAL EXAMINATION

The physical examination in the setting of hip dysplasia can be very similar to that of FAI in many patients. Pain with resisted straight and impingement tests is a marker of intra-articular hip pain, rather than being specific to FAI. Assessment of any abnormalities of gait, including alterations in foot-progression angle, are important to observe. The Trendelenburg sign is an important assessment of hip abductor weakness and is commonly positive in the dysplastic hip, although not specific to this pathology. Supine hip range of motion (ROM) should include assessment of hip flexion, abduction, and adduction, as well as rotational motion in 90 degrees of flexion (internal rotation in flexion [IRF], external rotation in flexion [ERF]). Increased IRF is generally a marker of underlying increased femoral anteversion, rather than the acetabular deformity. Decreased IRF or normal IRF (20 to 30 degrees) is present in many dysplastic hips with associated head-neck junction deformity or normal femoral version and does not rule out associated hip dysplasia. The apprehension test is a useful ancillary test for symptomatic anterior hip instability. This test is performed supine with the hip extended and then the hip is externally rotated. This can easily

be accomplished over the side of the exam table or by moving the patient down to the end of the exam table. Pain or apprehension in this position should suggest underlying hip instability. Assessment of internal (psoas) hip snapping is also useful because this is commonly present in the setting of hip flexor overload. Prone examination of internal and external rotation and the Craig test (trochanteric prominence angle) are helpful to provide a clinical estimate of femoral version. Diagnostic hip injections performed under fluoroscopy or ultrasound can serve as a useful extension of the physical examination. Intra-articular injection of anesthetic followed by a repeat physical examination can be useful to localize pain to an intra-articular source.

IMAGING

Traditional classifications of hip dysplasia have focused on differentiating the relatively severe forms of hip dysplasia. The Crowe radiographic classification has classically been used to describe the severity of hip dysplasia, with grade 1 hips having superior subluxation of the femoral head less than 50%, grade 2 hips having subluxation 50% to 75%, grade 3 hips having subluxation 75% to 100%, and grade 4 hips having subluxation greater than 100%.²⁰ Hartofilakidis provided alternative classification as dysplastic, low dislocation, or high dislocation.²¹ A dysplastic hip remains contained within the original acetabulum with varying degrees of subluxation. In low dislocations, there is overlap between the true and false acetabulum, whereas in high dislocations the femoral head is completely outside of the true acetabulum.

Although severe forms of hip dysplasia are commonly recognized at birth or in early childhood, milder forms of acetabular dysplasia commonly present in adolescence or young adulthood. A detailed assessment of the hip imaging is important to recognize milder forms of hip dysplasia. The anteroposterior (AP) pelvis radiograph is the standard for diagnosis of hip dysplasia and represents the historical standard used in studies investigating the natural history of hip dysplasia.²² The AP pelvis radiograph is preferred over AP hip views, due to the differing appearance of acetabular coverage and version between these views due to the beam location. Recently, many clinicians have favored the standing AP pelvis over supine AP pelvis radiograph because it represents the functional position of the pelvis. Pelvic tilt is commonly variable in this population, with some patients demonstrating anterior pelvic tilt that may allow for more functional acetabular coverage.²³

The LCEA is the most commonly used radiographic parameter of acetabular dysplasia. Measurement of the LCEA is fairly straightforward when done systematically and has been shown to be highly reliable.²⁴ Accurate measurement of the LCEA requires identification of the femoral head center, identification of the horizontal reference of the pelvis (using ischial tuberosities or teardrops), and identification of the lateral extent of the acetabular sourcil. The extent of the congruent acetabular sourcil should be used for measurement of the LCEA, rather than the most lateral extent of the entire acetabulum, which may overestimate the LCEA. LCEA measurements less than 20 degrees are generally defined as acetabular dysplasia, with measurements of 15 to 20 degrees representing mild acetabular dysplasia. LCEA values of

20 to 25 degrees define borderline acetabular dysplasia. Measurement of the acetabular inclination, or Tönnis angle, is also a useful adjunct to the LCEA and quantifies the obliquity of the weight-bearing acetabular surface. AI measurements greater than 15 degrees are indicative of dysplasia, whereas values of 10 to 15 degrees are viewed as borderline. Lateralization of the hip joint center is also commonly present even without hip subluxation and places additional stress of the abductor musculature. Assessment of an acetabular version on AP pelvis radiographs should be performed. Verification of normal pelvic rotation is important prior to assessing acetabular version. Similarly, an understanding of the underlying pelvic tilt is also important to take into context because it also alters the appearance of the acetabular rims. Acetabular retroversion has been shown to be present in one in six hips with acetabular dysplasia and appears to be more common in males.

The false profile radiograph was originally described by Lequesne as a useful assessment of anterior coverage of the acetabulum. The false profile radiograph is performed in the upright position with 65 degrees of pelvic rotation compared with the AP pelvis radiograph. The anterior center edge angle (ACEA), similar to the LCEA, is measured on this radiograph, with values less than 20 degrees indicating dysplasia. The relevance of head-neck offset deformities and asphericity is increasingly recognized in cases of combined FAI and dysplasia, as well as for FAI occurring after PAO (Fig. 81.3). Commonly used lateral radiographs most commonly include the 45-degree Dunn view and frog lateral radiographs. Assessment of head-neck junction morphology, including the alpha angle and head-neck offset ratio, is important in hip dysplasia. In addition, correction (or overcorrection) of the acetabular bony deformity may create FAI. The recognition of FAI after PAO was an important point that allowed Ganz and colleagues to refine the concept of FAI.

Magnetic resonance imaging (MRI) plays an important role in the assessment of intra-articular pathology in the setting of hip dysplasia. Similar to in other young adult hip disorders, MR arthrography (MRA) may optimally image the acetabular labrum and cartilage, although 3.0 Tesla scans may minimize the advantage of MRA. In the setting of hip dysplasia, the morphology of the acetabular labrum and its degree of hypertrophy are important to note. The hypertrophic labrum will commonly demonstrate significant intrasubstance degeneration in addition to detachment. The anterosuperior acetabular rim should be scrutinized for articular cartilage changes and associated acetabular rim edema or cyst formation. Similarly, edema in the femoral head is a worrisome finding that may indicate progressive end-stage disease. Particularly in the setting of hip dysplasia, delayed gadolinium-enhanced magnetic resonance imaging of cartilage may improve the assessment of the biochemical quality of articular cartilage compared with referenced standards.

CT represents the “gold standard” for assessment of bony morphology. However, radiation exposure from CT has traditionally limited the utilization of this technology to otherwise young, healthy populations. The development of low-dose CT protocols for hip imaging now allows for more routine utilization of low-dose CT for preoperative planning with a radiation exposure similar to three to four AP pelvis radiographs.²⁵ These

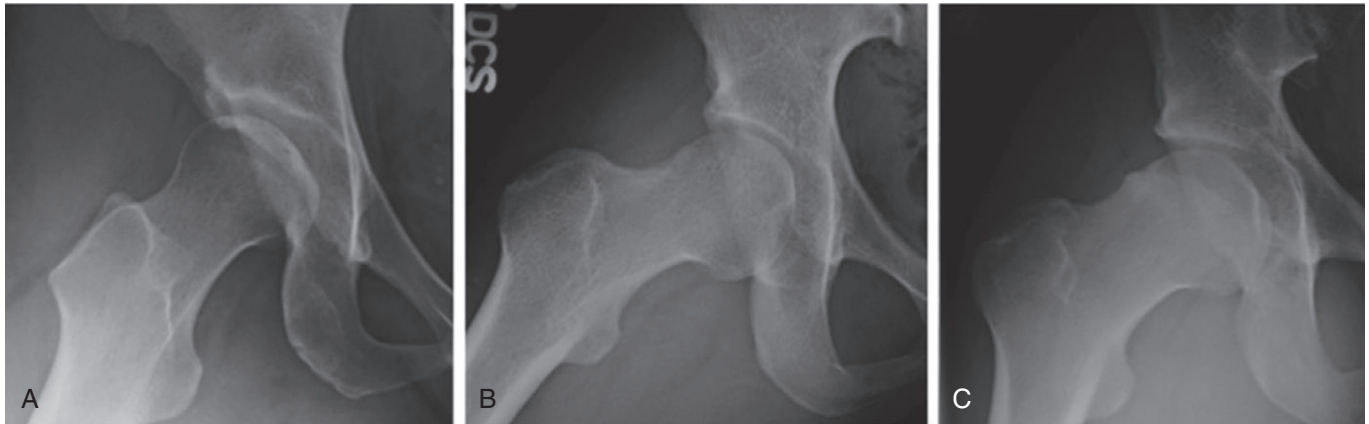


Fig. 81.3 Femoral morphology variants in hip dysplasia. (A) Dysplastic femoral morphology with associated asphericity. (B) Mildly decreased head-neck offset. (C) Severe cam morphology with convexity of head-neck junction in setting of acetabular dysplasia. (Reprinted from Nepple JJ, Clohisy JC. The dysplastic and unstable hip: a responsible balance of arthroscopic and open approaches. *Sports Med Arthrosc.* 2015;23[4]:180–186.)

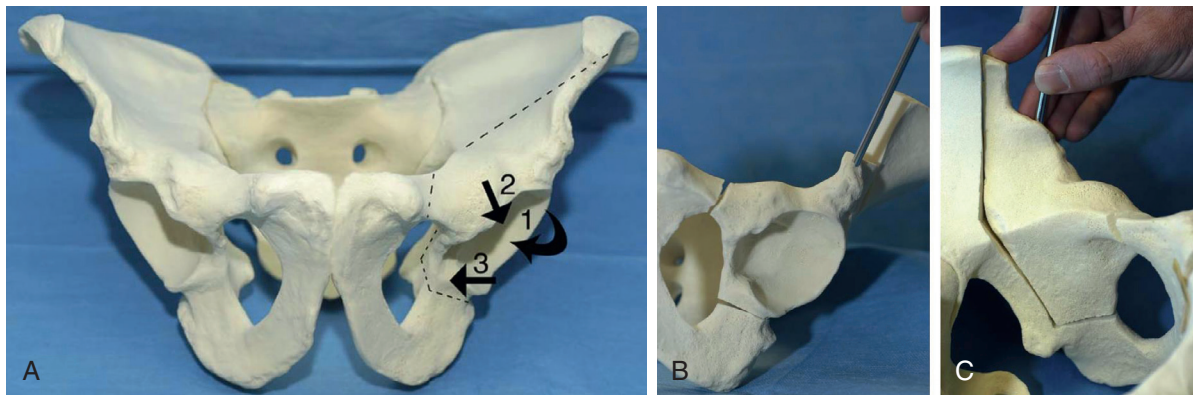


Fig. 81.4 Sawbones model of periacetabular cuts and acetabular reduction of dysplastic hip with PAO. The goal of reorientation is to enhance anterolateral femoral head coverage, to maintain or obtain acetabular anteversion, and to translate the hip center medially if indicated. (A) We perform the acetabular reduction with (1) internal rotation (lateral coverage and anteversion), (2) forward tilt or extension (anterior coverage), and (3) medial translation (medialization of joint center). (B, C) Periacetabular cuts are demonstrated. (Reprinted from Clohisy JC, Barrett SE, Gordon JE, Delgado ED, Schoenecker PL. Periacetabular osteotomy in the treatment of severe acetabular dysplasia. Surgical technique. *J Bone Joint Surg.* 2006;88[suppl 1, Pt 1]:65–83.)

low-dose CT protocols limit radiation exposure through optimization of dosing parameters (kV, mAs) and limiting the field of view (inferior sacroiliac joint to the lesser trochanters). Three-dimensional assessment via low-dose CT is increasingly used in the setting of hip dysplasia and can be particularly useful in the setting of complex or borderline deformity. Comprehensive characterization of the underlying deformity may play a role in improving the precision and accuracy of acetabular correction.

DECISION-MAKING

Hip Dysplasia

Surgical treatment of hip dysplasia most commonly involves correction of acetabular dysplasia through acetabular reorientation. A variety of pelvic osteotomies has been historically used in this population. In 1988 Ganz described the technique for a new Bernese PAO.²⁶ The PAO uses a modified Smith-Petersen approach to perform cuts of the ischium, pubis, ilium, and posterior column (Fig. 81.4). The posterior column cuts preserve

integrity of the posterior column and provides additional stability to the osteotomy. Since that time the PAO has become the gold-standard in North America for treatment of acetabular dysplasia in the skeletally mature patient. PAO techniques continue to evolve and minimize the recovery from this surgical procedure. Rectus-sparing techniques not requiring transection and repair of the direct head of the rectus tendon are now routinely used. Large multicenter cohort studies have demonstrated low complication rates of the PAO in the hands of experienced surgeon. Zaltz et al.²⁷ reported a major complication rate of 5.9% (grade 3 or 4 modified Dindo-Clavien classification) in the Academic Network of Conservation Hip Outcomes Research (ANCHOR) PAO cohort. Acetabular correction likely plays an important role in ultimate outcome of PAO, but the optimal outcome remains to be seen (Fig. 81.5). Recent studies have established normative data for acetabular coverage based on three-dimensional imaging.²⁸ Acetabular correction with residual anteversion or lateral under-coverage would tend to lead to residual hip instability, whereas acetabular correction with acetabular retroversion or lateral

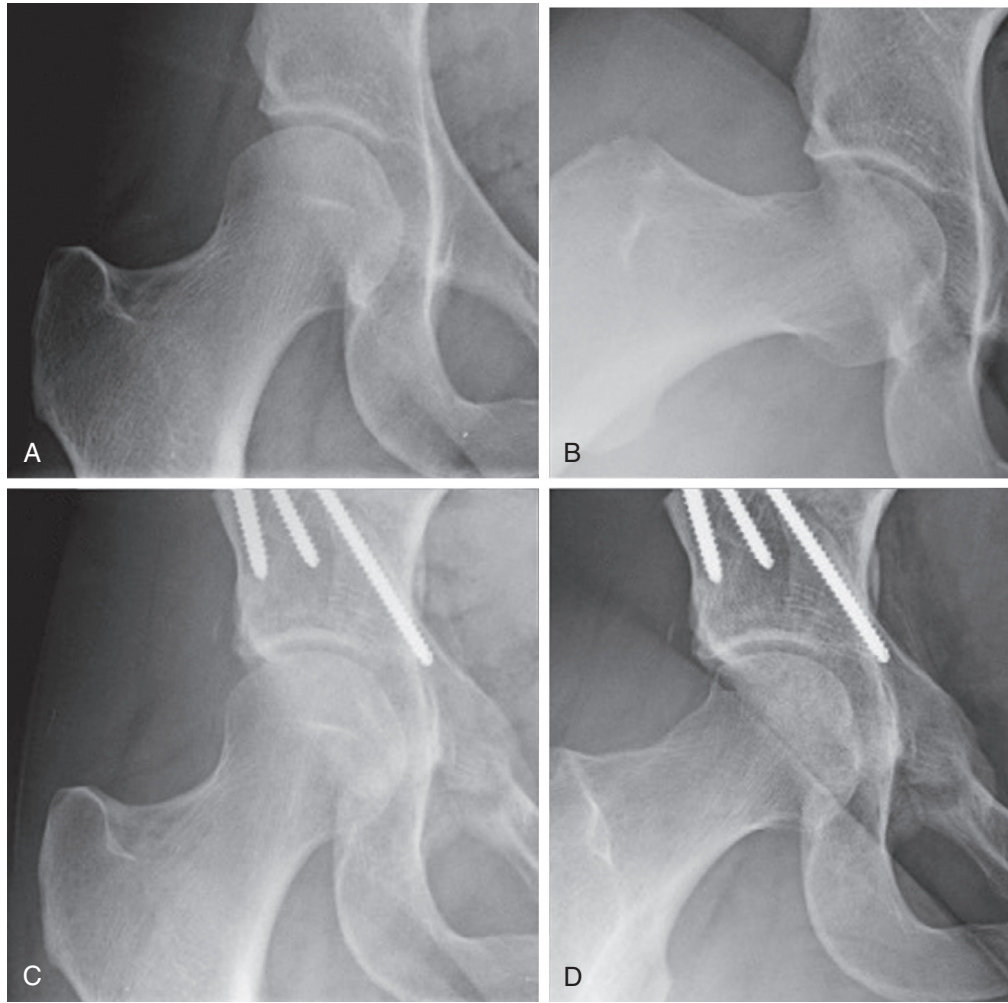


Fig. 81.5 Case example of mild/borderline dysplasia treated with periacetabular osteotomy (PAO). (A and B) Preoperative radiographs of a 23-year-old female patient with persistent right hip pain after previous hip arthroscopy with labral débridement, attempted osteoplasty, and unrepaired capsulotomy; (C and D) Post-operative radiographs 1 year after PAO with correction of underlying dysplasia and excellent clinical outcome.

overcoverage would tend to lead to FAI and ROM limitations. Fluoroscopic or plain radiographic assessment of correction before and after final fixation of the PAO is critical to optimize correction. It is important to ensure a horizontal reference of the pelvis is correct. In addition, it is important to understand the potential differences in acetabular wall projections between a fluoroscopic AP hip view and an AP pelvis view. The AP hip view will tend to make the acetabulum appear more anteverted than the AP pelvis view.

PAO is now commonly combined with other procedures in many patients in an attempt to optimize outcome. Significant labral and chondral pathology appears to be present in many, but not all, symptomatic dysplastic hips. Ross et al.²⁹ reported greater than 60% rates of labral and acetabular cartilage pathology requiring treatment in an early cohort undergoing hip arthroscopy and PAO. Increasing dysplasia severity was associated with more severe intra-articular pathology. Younger patients earlier in the disease pathophysiology may have pain secondary to dysplasia without developing structural damage to the labrum or articular cartilage. On the other hand, treatment of labral and chondral

pathology in addition to PAO may have a role in optimizing the outcomes of PAO. MRI (or MRA) is routinely used to screen for significant labrochondral pathology. In addition, the presence of sharp groin pain and mechanical symptoms should also raise concern for underlying intra-articular pathology. In cases of mild to moderate dysplasia, hip arthroscopy has a growing role in combination with PAO. Hip arthroscopy can be safely performed in the same setting as PAO. An efficient, central compartment arthroscopy can deal with labral and chondral pathology, while minimizing soft tissue extravasation of fluid that can make the PAO exposure more difficult. Although peripheral compartment work can be performed arthroscopically, we have found it more efficient to perform osteoplasty through the open incision. This approach minimizes additional fluid extravasation during this portion of the procedure and also allows for performance of the osteoplasty after the PAO correction. Hip arthroscopy combined with PAO in this manner appears to demonstrate no significant increase in complication rates compared with PAO alone. Performing the osteoplasty after the PAO correction allows for head-neck offset recontouring to be adjusted depending on the

ROM and minimize the risk of FAI after PAO. In the setting of severe acetabular dysplasia, hip arthroscopy can be more challenging and we prefer open arthrotomy to treat labral pathology.

The role of hip arthroscopy in the setting of acetabular dysplasia has evolved significantly over the past decade. In general, the use of hip arthroscopy for the treatment of young adult hip disorders has expanded rapidly during this period. Early reports of hip arthroscopy in the setting of acetabular dysplasia were encouraging. Byrd et al.³⁰ reported good results (80% with significant clinical improvement) in a cohort of 16 dysplastic hips and 32 borderline dysplastic hips at 2 years postoperatively (two patients undergoing total hip arthroplasty). However, subsequent reports demonstrated significant complications and inferior outcomes. Parvizi et al.³¹ reported significant complications, including postoperative dislocation and rapid progression to osteoarthritis in a cohort of 34 patients. Larson et al.³² reported a 32% failure rate among a series of 88 dysplastic or borderline dysplastic hips. With modern techniques including labral repair and capsular plication, the failure rate decreased to 18% but remains relatively high at short-term follow-up.

The midterm to long-term outcomes have been reported on the early experience of the PAO. At 20 years after PAO, survivorship is approximately 60%.³³ Compared with the available natural history studies, this would suggest an improvement associated with the PAO, although further studies will continue to establish the role of PAO in altering the natural history. The ANCHOR prospective multicenter cohort of PAO reported 2-year outcomes of 391 hips undergoing PAO, including 93% satisfaction with the procedure and major improvements in modified Harris hip score (23.6 point) and hip disability and osteoarthritis outcome score subscales (>20 points on all subscales including pain, activities of daily living, and sports/recreation).³⁴ In addition, data from our group document excellent long-term survivorship of 92% at 15 years.⁴⁴

Several factors have been associated with inferior outcomes after PAO in these early cohorts. Advanced articular cartilage damage has been associated with inferior outcomes.³³ A post-PAO lateral center edge angle less than 30 or greater than 40 degrees has also been associated with poorer outcome.³⁵ Other studies have supported undercorrection (including persistent lateralization, extrusion index >20%, excessive acetabular anteversion, and lateral center edge angle <22 degrees) to be associated with poorer outcome.^{33,36,37} FAI after acetabular correction remains to be better defined.^{37,38} Albers et al.³⁷ have demonstrated inferior outcomes and disease progression after PAO when FAI occurs. This included acetabular overcoverage, acetabular retroversion, and lack of head-neck junction osteoplasty in the setting of an aspherical femoral head-neck junction. A detailed assessment of ROM after final fixation of the PAO is important. Satisfactory ROM is generally considered at least 90 degrees of hip flexion and at least 15 degrees of IRE. Consideration of the functional ROM requirements of athletes is also important. In the setting of limited ROM an osteoplasty performed through an open arthrotomy allows for correction of FAI. The anterior inferior iliac spine has been increasingly recognized as a potential source of subspine impingement, which is also relevant to the post-PAO hip. The long-term effect of osteoplasty of outcomes after PAO remains to be seen.

Borderline Dysplasia

Borderline hip dysplasia is a commonly but poorly investigated condition. Several natural history studies indicate that the risk of osteoarthritis in hip dysplasia may extend up to an LCEA of 25 degrees or potentially higher. Borderline hip dysplasia is a commonly used term to refer to patients with LCEA between 20 and 25 degrees. Patients in this subgroup may have symptomatic hip instability, whereas others appear to have symptoms secondary to underlying FAI, or in some cases both instability and impingement. It is important to recognize that measurements in the borderline range are more common than those in the dysplastic range. For the female athlete, particularly in a high hip ROM sport, borderline acetabular coverage is extremely common. In a cross-sectional study of female collegiate athletes, 21% had evidence of dysplasia (LCEA <20 degrees) and an additional 46% had borderline dysplasia (LCEA 20 to 25 degrees).³⁹ In total, 67% had an LCEA of 25 degrees or less. In a similar study in male collegiate athletes, 7% demonstrated dysplasia and an additional 19% borderline dysplasia.⁴⁰ Together these studies emphasize that borderline acetabular dysplasia is present in approximately three times as many hips as true acetabular dysplasia. In this population, clinical decision-making remains challenging and requires a thorough understanding of all relevant patient-specific factors to choose the best treatment. Important factors in this population include sex, age, soft tissue laxity (Beighton score), ROM, femoral version, and head-neck morphology. Three-dimensional characterization of acetabular morphology may provide additional information in borderline cases to determine the optimal treatment.

The role of hip arthroscopy versus PAO in the setting of borderline acetabular dysplasia remains controversial. As previously mentioned, Larson et al. reported relatively high rates (greater than 30%) of failure of arthroscopy in the setting of dysplasia or borderline dysplasia.³² Domb et al.⁴¹ reported the outcomes of 22 patients with borderline acetabular dysplasia (study definition of 18 to 25 degrees) treated with hip arthroscopy and capsular plication at 2 years postoperatively. In this study, good/excellent short-term outcomes were reported in 80% of patients (one patient suffered dislocation postoperatively; none required total hip arthroplasty). Nevertheless, larger studies with longer follow-up that analyze all characteristics of hip instability and impingement are needed to clarify the best treatments in the borderline hip.

Atraumatic Hip Microinstability

The role of soft tissue laxity in the hip continues to be better defined. The role of soft tissue laxity in other joints, including the shoulder and knee, is well established, and only recently has it received significant attention in the hip. Gross hip instability can occur without major trauma but is exceptionally rare. Atraumatic hip microinstability is currently recognized as a clinical disorder. Capsular plication has emerged as a potential treatment. Initially, open capsular plication was used, but as arthroscopic techniques have improved, arthroscopic capsular plication emerged as a common treatment technique. Capsule closure appears to improve early outcomes compared with unrepaired

capsulotomy in the setting of hip arthroscopy.⁴² Arthroscopic capsular plication techniques are generally performed similar to those of capsular closure with larger bites of the capsular flaps used or with resection of some of the distal capsular flap. Capsular plication with an oblique capsular (distal medial to proximal lateral) shift has been advocated to maximize the effect of the plication.⁴¹

Early reports of the role of capsular plication in selected patients have been encouraging. Domb et al.⁴¹ reported good results of capsular plication as part of the arthroscopic treatment

of a cohort of patients with borderline hip dysplasia. Larson et al.⁴³ reported excellent outcomes of hip arthroscopy with capsular plication in a cohort of females with Ehlers-Danlos syndrome. Future studies will be important to determine the generalizability of these results. The longevity of capsular plication remains a concern, because recurrent instability from stretching out of plication appears to be an issue with other joint plications such as the shoulder.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

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